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# Low-Complexity Estimation for Conditional Outage Prediction in Nakagami-*m* MRC Systems

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***Abstract*—In this letter, we propose a three-parameter bivariate Nakagami-$m$ model for conditional outage prediction in maximal-ratio combining (MRC) systems. The model has triangular support and jointly describes an observed branch SNR and the post-combining SNR. Parameter estimation is performed using a profiled maximum likelihood estimator, a moment estimator, and a closed-form modified-likelihood estimator obtained from algebraic estimating equations. The simulations show that the closed-form estimator improves on the moment estimator for $m_1$ and is substantially faster than the MLE, which remains more accurate for $m_2$. Sensitivity to initialization, unequal powers, and correlated branches is also evaluated.**

***Index Terms*—Nakagami-$m$ fading, maximal-ratio combining, conditional outage prediction, low-complexity estimation, closed-form estimation.**

## I. Introduction

CONDITIONAL outage prediction is useful in pilot-assisted receivers when a branch-level signal-to-noise ratio (SNR) is observed before the post-combining SNR is available. The conditional probability that the maximal-ratio combining (MRC) output is below a threshold can then be used for link adaptation and branch selection. The Nakagami-$m$ law is widely used to model fading because of its flexibility and analytical tractability. Let $X$ be Nakagami-$m$. Its density is

$$f(x;m,\Omega) = \frac{2}{\Gamma(m)}\left(\frac{m}{\Omega}\right)^m x^{2m-1}\exp\left(-\frac{m}{\Omega}x^2\right),$$

where $m \geq \frac{1}{2}$ controls fading severity and $\Omega > 0$ is the average power [1]. While the univariate Nakagami-$m$ model is well suited for marginal fading characterization, receiver-level tasks such as maximal-ratio combining (MRC) naturally involve *joint* statistics of related branch and post-combining powers.

Bivariate fading models have been used to describe joint envelope behavior in multi-branch systems. Several bivariate Nakagami-$m$ formulations have been proposed to model correlated envelopes in performance analyses [2], [3]. However, some constructions involve special functions, infinite-series representations, or parameter restrictions that complicate statistical estimation. In these cases, the estimators are obtained by numerical optimization, which increases the computational time when the parameters are updated repeatedly.

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In this letter, we develop a low-complexity estimation approach for branch-conditioned outage prediction in Nakagami-$m$ MRC systems. To this end, we propose a three-parameter bivariate Nakagami-$m$ model, in which one component represents a branch-level SNR contribution and the other represents the corresponding post-combining SNR. The proposed model has triangular support, closed-form Nakagami marginals, and a tractable conditional distribution. These properties allow direct evaluation of conditional outage probabilities.

For inference, we consider the standard MLE with profiled scale, a moment-based estimator, and a closed-form estimator derived from a structural specialization of the likelihood [4], [5]. CMLE denotes the *closed-form modified-likelihood estimator*, defined by modified score equations from the encompassing Gamma formulation; it is an estimating-equation estimator distinct from the exact three-parameter MLE. The proposed estimator does not require numerical maximization and can be used when the parameters must be updated repeatedly. Related closed-form estimators for bivariate and multivariate gamma models are known to be strongly consistent and asymptotically normal, and, in the multivariate case, to share the same first-order asymptotic behavior as the MLE [6], [7].

The remainder of this letter is organized as follows. Section II introduces the proposed bivariate Nakagami-$m$ distribution and its key properties. Section III presents the MLE and the method of moments, while Section IV develops the closed-form estimator. Section V reports Monte Carlo results. Section VI illustrates the model through conditional outage prediction in dual-branch MRC. Section VII concludes.

## II. Bivariate Nakagami-*m* Distribution

We introduce a three–parameter bivariate Nakagami-$m$ model for positive pairs $(Y_1, Y_2)$ supported on $\{(y_1, y_2) : 0 < y_1 < y_2 < \infty\}$. Although some works allow arbitrary positive Nakagami parameters, in this letter we adopt the conventional fading range $m_i \geq 1/2$ used in the original Nakagami formulation and standard fading-channel treatments [1], [8]. The lower bound is a conventional physical fading restriction; the proposed density is mathematically valid for positive shapes. Let $m_1$ and $m_2$ be the shape parameters and let $\Omega > 0$ be a scale (power) parameter. The joint density is defined as

$$\begin{aligned} f_Y(y_1,y_2) = &\frac{4\,(m_1+m_2)^{m_1+m_2}}{\Gamma(m_1)\,\Gamma(m_2)\,\Omega^{m_1+m_2}}\; y_1^{2m_1-1}\left(y_2^2-y_1^2\right)^{m_2-1} \\ &\times\; y_2\,\exp\left(-\frac{(m_1+m_2)\,y_2^2}{\Omega}\right), \end{aligned} \tag{1}$$

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for $0 < y_1 < y_2 < \infty$. We write $(Y_1, Y_2) \sim \mathrm{BNak}(m_1, m_2, \Omega)$.

Throughout, we will use the quadratic transforms $Z_1 := Y_1^2$, $Z_2 := Y_2^2$, $0 < Z_1 < Z_2$, which will be convenient for likelihood calculations and simulation.

*Notation.* The shapes of the two Gamma increments are $m_1$ and $m_2$; $\beta = \Omega/(m_1+m_2)$ is their common Gamma scale; $Z_1$ is the observed branch SNR and $Z_2$ the accumulated MRC SNR; $n$ is the number of paired pilot observations, $R$ the number of Monte Carlo replications, and $\gamma_{\mathrm{th}}$ the outage threshold.

*Construction note.* The model admits the representation $V_1 \sim \mathrm{Ga}(m_1, \beta)$ and $V_2 \sim \mathrm{Ga}(m_2, \beta)$ independently, with $(Y_1, Y_2) = (\sqrt{V_1}, \sqrt{V_1 + V_2})$. This representation is related to the classical Beta–Gamma factorization [9], [10]. The Nakagami parameterization provides the triangular branch-to-MRC-output interpretation and is used below to obtain the CMLE and the conditional outage probability.

**Proposition II.1** (Marginals and conditional structure)**.** *If* $(Y_1, Y_2) \sim \mathrm{BNak}(m_1, m_2, \Omega)$ *as in* (1)*, then:*

*(a) The second component has Nakagami marginal with aggregated shape:*

$$Y_2 \sim \mathrm{Nakagami}(m_1 + m_2,\ \Omega), \quad \textit{i.e.}$$

$$f_{Y_2}(y_2) = \frac{2\,(m_1+m_2)^{\,m_1+m_2}}{\Gamma(m_1+m_2)\,\Omega^{\,m_1+m_2}}\, y_2^{\,2(m_1+m_2)-1} \times \exp\left(-\frac{(m_1+m_2)\,y_2^2}{\Omega}\right).$$

*In particular,* $\mathbb{E}[Y_2^2] = \Omega$.

*(b) The first component has Nakagami marginal with the same shape* $m_1$ *and an induced scale*

$$Y_1 \sim \mathrm{Nakagami}(m_1,\ \Omega_1), \quad \textit{where} \quad \Omega_1 := \frac{m_1}{m_1+m_2}\,\Omega,$$

*so that*

$$f_{Y_1}(y_1) = \frac{2\,m_1^{\,m_1}}{\Gamma(m_1)\,\Omega_1^{\,m_1}}\, y_1^{\,2m_1-1}\, \exp\left(-\frac{m_1\,y_1^2}{\Omega_1}\right),$$

$$\mathbb{E}[Y_1^2] = \Omega_1 = \frac{m_1}{m_1+m_2}\,\Omega.$$

*(c) Define the squared ratio* $U := (Y_1/Y_2)^2 \in (0,1)$*. Then*

$$U \sim \mathrm{Beta}(m_1, m_2) \quad \textit{and is independent of } Y_2^2.$$

*Equivalently, the conditional density of* $Y_1$ *given* $Y_2 = y_2$ *is*

$$f_{Y_1|Y_2}(y_1 \mid y_2) = \frac{2\,y_1\,\left(y_2^2 - y_1^2\right)^{m_2-1}\, y_1^{\,2m_1-2}}{y_2^{\,2(m_1+m_2)-1}\,\mathrm{B}(m_1, m_2)},$$

*where* $0 < y_1 < y_2$*, with* $\mathrm{B}(\cdot,\cdot)$ *the Beta function.*

*Proof.* Integrating (1) over $y_1 \in (0, y_2)$ with the substitution $u = (y_1/y_2)^2$ yields *(a)* via the Beta integral; integrating over $y_2 \in (y_1, \infty)$ gives *(b)* with induced scale $\Omega_1$. For *(c)*, the change of variables $(Y_1, Y_2) \mapsto (U, Y_2)$ with Jacobian $2y_2^2 u^{1/2}$ factorizes the joint density, showing $U \perp Y_2^2$ and $U \sim \mathrm{Beta}(m_1, m_2)$. □

It is worth mentioning that the model captures positive dependence through the Beta law of $U = (Y_1/Y_2)^2$; larger $m_2$ concentrates mass away from the boundary $y_1 \approx y_2$. Additionally, the parameterization is identifiable: $\Omega = \mathbb{E}[Y_2^2]$ and $\Omega_1/\Omega = m_1/(m_1+m_2)$, while the $\mathrm{Beta}(m_1, m_2)$ law of $U$ identifies the shape parameters $(m_1, m_2)$.

Additionally, the Pearson correlation of the *squared* envelopes admits the closed form

$$\rho(Z_1, Z_2) = \sqrt{\frac{m_1}{m_1+m_2}} \in (0,1),$$

which depends only on $(m_1, m_2)$ (it is scale-free in $\Omega$).

## III. Estimation

Let $\{(y_{1j}, y_{2j})\}_{j=1}^n$ be i.i.d. from the bivariate Nakagami-$m$ model in (1) with support $0 < y_{1j} < y_{2j} < \infty$. We adopt the quadratic transforms:

$$z_{1j} := y_{1j}^2, \qquad z_{2j} := y_{2j}^2, \qquad 0 < z_{1j} < z_{2j}.$$

Define the convenient summaries

$$S_{\log 1} := \sum_{j=1}^n \log z_{1j}, \quad S_{\log 2} := \sum_{j=1}^n \log z_{2j},$$

$$S_{\log \Delta} := \sum_{j=1}^n \log(z_{2j} - z_{1j}), \quad S_2 := \sum_{j=1}^n z_{2j}.$$

From (1), the log-likelihood for $\theta = (m_1, m_2, \Omega)$ is

$$\begin{aligned}\ell(\theta) =& n\big[\log 4 + (m_1+m_2)\log(m_1+m_2) - (m_1+m_2)\log\Omega \\ & - \log\Gamma(m_1) - \log\Gamma(m_2)\big] + \Big(m_1 - \tfrac{1}{2}\Big) S_{\log 1} \\ & + (m_2 - 1) S_{\log\Delta} + \tfrac{1}{2} S_{\log 2} - \frac{m_1+m_2}{\Omega}\, S_2.\end{aligned} \tag{2}$$

Profiling w.r.t. $\Omega$ (i.e., setting $\partial\ell/\partial\Omega = 0$ for fixed $m_1, m_2$) yields the closed-form

$$\widehat{\Omega}_{\mathrm{prof}}(m_1, m_2) = \frac{S_2}{n}, \tag{3}$$

so, the MLE reduces to maximizing the log-likelihood of the two-parameter profile $\ell_p(m_1, m_2) = \ell\big(m_1, m_2, \widehat{\Omega}_{\mathrm{prof}}(m_1, m_2)\big)$, which can be done with standard Newton or quasi-Newton routines (the scores involve digamma terms from $\partial_{m_i} \log\Gamma(m_i)$). Regularity requires $m_i \geq 1/2$ and $z_{2j} > z_{1j}$.

**Method of Moments (MoM).** The model implies $Z_2 = Y_2^2$ has mean $\mathbb{E}[Z_2] = \Omega$, and the squared ratio

$$U := \left(\frac{Y_1}{Y_2}\right)^2 = \frac{Z_1}{Z_2} \in (0,1)$$

is $\mathrm{Beta}(m_1, m_2)$ and independent of $Z_2$ (cf. Prop. II.1). Let

$$\overline{Z}_2 := \frac{1}{n}\sum_{j=1}^n z_{2j}, \quad u_j := \frac{z_{1j}}{z_{2j}},$$

$$\overline{U} := \frac{1}{n}\sum_{j=1}^n u_j, \quad s_U^2 := \frac{1}{n}\sum_{j=1}^n (u_j - \overline{U})^2.$$

Matching the first two moments of a $\mathrm{Beta}(\alpha, \beta)$ law ($\mu = \alpha/(\alpha+\beta)$, $\sigma^2 = \mu(1-\mu)/(\alpha+\beta+1)$) gives

$$\widehat{\kappa}_{\mathrm{MoM}} := \frac{\overline{U}(1-\overline{U})}{s_U^2} - 1, \qquad \widehat{m}_1 = \overline{U}\,\widehat{\kappa}_{\mathrm{MoM}}, \tag{4}$$

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$$\widetilde{m}_2 = (1-\overline{U})\,\widehat{\kappa}_{\mathrm{MoM}}, \qquad \widetilde{\Omega} = \overline{Z}_2. \tag{5}$$

Feasibility requires $0 < \overline{U} < 1$ and $s_U^2 < \overline{U}(1-\overline{U})$ so that $\widehat{\kappa}_{\mathrm{MoM}} > 0$; in samples with highly concentrated $u_j$, a slight ridge on $s_U^2$ stabilizes (eq. 5).

The MLE in (2) requires numerical optimization over $(m_1, m_2)$. The MoM in (5) has a closed-form expression but may be less efficient for moderate sample sizes. Therefore, we consider a structural specialization of the likelihood that provides closed-form estimators for $(m_1, m_2)$, referred to as CMLE, while retaining the profiled scale in (8).

## IV. Closed-Form Estimators

To obtain algebraic estimating equations, we use a power-transformed version of the bivariate Gamma partial-sum construction [10]. On $\mathcal{S}_{\boldsymbol{\gamma}} = \{(y_1, y_2) > 0 : y_1^{\gamma_1} < y_2^{\gamma_2}\}$, consider parameters $(\alpha_1, \alpha_2, \beta) > 0$ and exponents $\boldsymbol{\gamma} = (\gamma_1, \gamma_2) > 0$. The resulting generalized bivariate Gamma (GBG) density is

$$g(y_1, y_2 \,|\theta, \gamma_1, \gamma_2) = \frac{\gamma_1\gamma_2}{\beta^{\alpha_1+\alpha_2}\Gamma(\alpha_1)\Gamma(\alpha_2)}\, y_1^{\alpha_1\gamma_1-1} \times \left(y_2^{\gamma_2} - y_1^{\gamma_1}\right)^{\alpha_2-1} y_2^{\gamma_2-1} \exp\Big(-\frac{y_2^{\gamma_2}}{\beta}\Big). \tag{6}$$

where $\theta = (\alpha_1, \alpha_2, \beta)$. For an i.i.d. sample $\{(y_{1j}, y_{2j})\}_{j=1}^n$ the log-likelihood reads

$$\begin{aligned}\ell(\theta, \boldsymbol{\gamma}) = {} & n(\log\gamma_1 + \log\gamma_2) - (\alpha_1+\alpha_2)n\log\beta - n\log\Gamma(\alpha_1) \\ & + (\alpha_1\gamma_1 - 1)\sum_j \log y_{1j} + (\alpha_2 - 1)\sum_j \log\big(y_{2j}^{\gamma_2} - y_{1j}^{\gamma_1}\big) \\ & + (\gamma_2 - 1)\sum_{j=1}^n \log y_{2j} - n\log\Gamma(\alpha_2) - \frac{1}{\beta}\sum_{j=1}^n y_{2j}^{\gamma_2},\end{aligned} \tag{7}$$

The closed-form modified-likelihood estimator is obtained by differentiating the GBG log-likelihood in (7) with respect to $(\gamma_1, \gamma_2)$ and evaluating the derivatives at $\gamma_1 = \gamma_2 = 2$, $\alpha_1 = m_1$, $\alpha_2 = m_2$, and $\beta = \Omega/(m_1+m_2)$. This procedure defines estimating equations that differ from direct maximization of (2). Their validity under the BNak model follows from the moment identities presented below. Working with the quadratic transforms $z_{1j} = y_{1j}^2$, $z_{2j} = y_{2j}^2$ (with $0 < z_{1j} < z_{2j}$), the scale score $\partial\ell/\partial\beta = 0$ yields

$$\widehat{\Omega} = \frac{1}{n}\sum_{j=1}^n z_{2j}. \tag{8}$$

Introducing

$$S_{\log 1} := \sum_{j=1}^n \log z_{1j}, \quad S_{\log 2} := \sum_{j=1}^n \log z_{2j}, \quad S_2 := \sum_{j=1}^n z_{2j},$$

$$S_{\log\Delta} := \sum_{j=1}^n \log(z_{2j} - z_{1j}), \quad S_{2\log 2} := \sum_{j=1}^n z_{2j}\log z_{2j},$$

$$S_{\frac{1\log 1}{\Delta}} := \sum_{j=1}^n \frac{z_{1j}\log z_{1j}}{z_{2j} - z_{1j}}, \quad S_{\frac{2\log 2}{\Delta}} := \sum_{j=1}^n \frac{z_{2j}\log z_{2j}}{z_{2j} - z_{1j}},$$

the two modified exponent-score equations, after evaluating at $(\gamma_1, \gamma_2) = (2, 2)$, are

$$\begin{aligned}\psi_{1n}(m_1, m_2) &= n + m_1 S_{\log 1} - (m_2 - 1) S_{\frac{1\log 1}{\Delta}} = 0, \\ \psi_{2n}(m_1, m_2, \Omega) &= n + S_{\log 2} + (m_2 - 1) S_{\frac{2\log 2}{\Delta}} \\ &\quad - \frac{m_1 + m_2}{\Omega} S_{2\log 2} = 0.\end{aligned} \tag{9}$$

Together with the profiled scale, these equations form a linear system in $(m_1, m_2)$ and provide an algebraic solution. Straightforward algebra leads to the closed-form, MLE-like estimators (CMLE)

$$\widehat{m}_2 = -\frac{D}{C}, \qquad \widehat{\Omega} = \frac{S_2}{n},$$
$$\widehat{m}_1 = \frac{(\widehat{m}_2 - 1) S_{\frac{1\log 1}{\Delta}} - n}{S_{\log 1}}.$$

with

$$C := \frac{1}{n} S_{\frac{2\log 2}{\Delta}} S_2 S_{\log 1} - S_{\frac{1\log 1}{\Delta}} S_{2\log 2} - S_{\log 1} S_{2\log 2},$$

$$\begin{aligned}D := {} & S_{\frac{1\log 1}{\Delta}} S_{2\log 2} - \frac{1}{n} S_{\frac{2\log 2}{\Delta}} S_2 S_{\log 1} + n S_{2\log 2} \\ & + \Big(1 + \tfrac{1}{n} S_{\log 2}\Big) S_2 S_{\log 1}.\end{aligned}$$

**Proposition IV.1** (CMLE validity)**.** *Let $\theta_0 = (m_{10}, m_{20}, \Omega_0)$ generate the BNak model. If $m_{20} > 1$, $\mathbb{E}_{\theta_0}(\log Z_1) \neq 0$, and the population coefficient matrix associated with* (9) *is nonsingular, then the raw CMLE is consistent. If, additionally, $m_{20} > 2$, it is asymptotically normal with sandwich covariance.*

*Proof sketch.* Write $V_1 = Z_1$, $V_2 = Z_2 - Z_1$, $K = m_1 + m_2$, and $\beta = \Omega/K$. Gamma–Beta identities give

$$\mathbb{E}\frac{V_1 \log V_1}{V_2} = \frac{m_1\{\log\beta + \psi(m_1+1)\}}{m_2 - 1},$$
$$\mathbb{E}\frac{Z_2 \log Z_2}{V_2} = \frac{(K-1)\{\log\beta + \psi(K)\}}{m_2 - 1}.$$

Together with $\mathbb{E}\log V_1 = \log\beta + \psi(m_1)$, $\mathbb{E}\log Z_2 = \log\beta + \psi(K)$, and $\mathbb{E}(Z_2\log Z_2) = K\beta\{\log\beta + \psi(K+1)\}$, the recurrence $\psi(x+1) = \psi(x) + 1/x$ yields $\mathbb{E}\{\psi_{1n}(\theta_0)/n\} = \mathbb{E}\{\psi_{2n}(\theta_0)/n\} = 0$. Let $H_j$ collect the per-observation summaries in (9). The raw CMLE is a rational map $T(\bar{H}_n)$ of their sample means. For $m_{20} > 1$, the required expectations are finite and the identities above give $T\{\mathbb{E}(H_j)\} = \theta_0$. The nonzero denominator and nonsingularity conditions make $T$ continuous at $\mathbb{E}(H_j)$; hence the law of large numbers and the continuous mapping theorem give consistency. For $m_{20} > 2$, the inverse-gap terms have finite second moments, so the multivariate CLT and the Delta method give asymptotic normality.

The parameter space of the BNak distribution is $m_i \geq 1/2$; consistency and asymptotic normality of this CMLE apply under the respective shape conditions stated above.

*Finite-sample feasibility.* Because $-D/C$ may be nonpositive in finite samples, the raw estimate is used when $C$ is numerically separated from zero and both shapes are finite and at least $1/2$. Samples failing these conditions use an admissible MoM estimate or, when necessary, the constrained profiled MLE. The raw CMLE was admissible in all $10^4$ replications at every sample size of the reported design.

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**Asymptotic variance.** Let $\psi_j(\theta)$ denote the per-observation version of the two modified scores in (9), augmented by the scale equation, and define $A = \mathbb{E}\{-\partial\psi_j(\theta_0)/\partial\theta^\top\}$ and $B = \mathbb{E}\{\psi_j(\theta_0)\psi_j(\theta_0)^\top\}$. Under the regularity conditions above,

$$\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N_3\Big(0, A^{-1}BA^{-\top}\Big),$$

with a consistent sandwich estimate obtained by replacing $A$ and $B$ by their empirical counterparts. In particular, $\widehat{\Omega} = \bar{Z}_2$ and $\mathrm{Var}(\widehat{\Omega}) = \Omega^2/\{n(m_1+m_2)\}$. CMLE standard errors therefore follow from the estimating-equation sandwich matrix.

Writing $\psi_j(\theta) = \{\psi_{1j}(\theta), \psi_{2j}(\theta), Z_{2j} - \Omega\}^\top$, an empirical covariance estimator is

$$\widehat{\mathrm{Var}}(\widehat{\theta}) = \frac{1}{n}\widehat{A}^{-1}\widehat{B}\widehat{A}^{-\top},$$
$$\widehat{A} = -\frac{1}{n}\sum_{j=1}^{n}\frac{\partial\psi_j(\widehat{\theta})}{\partial\theta^\top},$$
$$\widehat{B} = \frac{1}{n}\sum_{j=1}^{n}\psi_j(\widehat{\theta})\psi_j(\widehat{\theta})^\top.$$

## V. Simulation Study

A Monte Carlo simulation study was performed to evaluate the finite-sample behavior of the profiled MLE, CMLE, and MoM. In each replication, an i.i.d. sample $(Y_1, Y_2)$ was generated from the proposed model using independent variables $V_1 \sim \mathrm{Ga}(m_1, \beta)$ and $V_2 \sim \mathrm{Ga}(m_2, \beta)$, with $(Z_1, Z_2) = (V_1, V_1 + V_2)$, $(Y_1, Y_2) = (\sqrt{Z_1}, \sqrt{Z_2})$, and $\beta = \Omega/(m_1 + m_2)$. We fixed $(m_1, m_2, \Omega) = (2, 3.5, 5)$, considered $n \in \{30, 60, 90, \ldots, 300\}$, and used $R = 10000$ replications for each sample size. One paired observation corresponds to one pilot snapshot; consequently, $n$ is the pilot-window size used for parameter estimation.

The MoM matches the first two moments of $U = Z_1/Z_2 \sim \mathrm{Beta}(m_1, m_2)$ and sets $\widehat{\Omega} = \bar{Z}_2$. The CMLE evaluates the closed-form expressions based on sample summaries involving $\log Z_1$, $\log Z_2$, $Z_2 \log Z_2$, and $(Z_i \log Z_i)/(Z_2 - Z_1)$. The profiled MLE was computed by numerically maximizing the profile log-likelihood in $(m_1, m_2)$, with $\widehat{\Omega} = \bar{Z}_2$, using MoM-based initial values and the constraint $m_i \geq 0.5$. For each $\theta \in \{m_1, m_2, \Omega\}$, we report Monte Carlo bias and mean squared error (MSE). Initialization sensitivity was evaluated from $(0.5, 0.5)$, $(1, 1)$, $(0.6, 10)$, $(10, 0.6)$, and $(20, 20)$ in 500 samples at $n \in \{30, 100, 300\}$. All five starts converged in at least 99.8% of samples, and the largest difference between converged shape estimates was $4.48\times10^{-4}$, indicating numerical stability over these starts.

Figure 1 shows that the MLE has the smallest bias and MSE. For $m_1$, the CMLE is close to the MLE and reduces the MSE relative to the MoM. For $m_2$, the CMLE and MoM have comparable MSE, whereas the MLE has a clear advantage. All methods use $\widehat{\Omega} = \bar{Z}_2$ and therefore share the same $\Omega$ curves.

## VI. Application to Conditional Outage Prediction in Dual-Branch MRC

We apply the proposed model to branch-conditioned outage prediction in a dual-branch MRC receiver. Consider a pilot-assisted system with two independently faded branches, and let

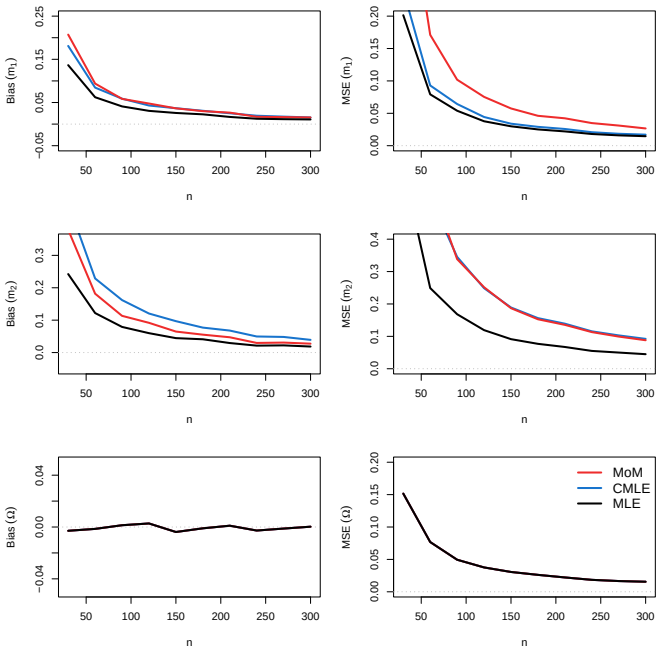



Fig. 1: Bias (left) and MSE (right) for $\widehat{m}_1$ (top), $\widehat{m}_2$ (middle), and $\widehat{\Omega}$ (bottom) under MoM (red), CMLE (blue), and MLE (black) for $n \in \{30, 60, \ldots, 300\}$, with $(m_1, m_2, \Omega) = (2, 3.5, 5)$ and $R = 10000$.

$\gamma_1$ and $\gamma_2$ denote the instantaneous branch SNR contributions. The MRC output SNR is $\gamma_{\mathrm{MRC}} = \gamma_1 + \gamma_2$. We define $Z_1 = \gamma_1$ and $Z_2 = \gamma_{\mathrm{MRC}}$, so that $0 < Z_1 < Z_2$. The two coordinates therefore describe successive SNR accumulation: a branch contribution and the MRC output. The common-scale assumption implies $\mathbb{E}(\gamma_i) = m_i\beta$ and is appropriate for power-normalized branches or links for which $\bar{\gamma}_i/m_i$ is approximately common. Distinct scales arising from unequal path losses fall outside the three-parameter model and are examined below as misspecification.

Unlike unconditional outage prediction, which depends only on the marginal law of $Z_2$, the conditional outage probability $P_{\mathrm{out}}(\gamma_{\mathrm{th}} \mid z_1) = \Pr(Z_2 < \gamma_{\mathrm{th}} \mid Z_1 = z_1)$ requires the bivariate structure. It quantifies the probability that the final MRC output falls below a target threshold given the currently observed branch-level SNR.

From the proposed construction, $Z_2 \mid Z_1 = z_1 \overset{d}{=} z_1 + W$, where $W \sim \mathrm{Ga}\{m_2, \Omega/(m_1 + m_2)\}$ under the shape–scale parameterization. Therefore,

$$P_{\mathrm{out}}(\gamma_{\mathrm{th}} \mid z_1) = \begin{cases} \dfrac{\gamma\Big(m_2, \frac{(m_1+m_2)(\gamma_{\mathrm{th}}-z_1)}{\Omega}\Big)}{\Gamma(m_2)}, & 0 < z_1 < \gamma_{\mathrm{th}}, \\ 0, & z_1 \geq \gamma_{\mathrm{th}}. \end{cases}$$

In practice, this curve is evaluated after replacing the unknown parameters by their estimates.

Figure 2 compares the true conditional outage curve with the fitted curves obtained from one reproducible pilot window of size $n = 200$, with $z_1$ fixed at its true median. All three fitted curves are close to the true curve over the threshold range considered.

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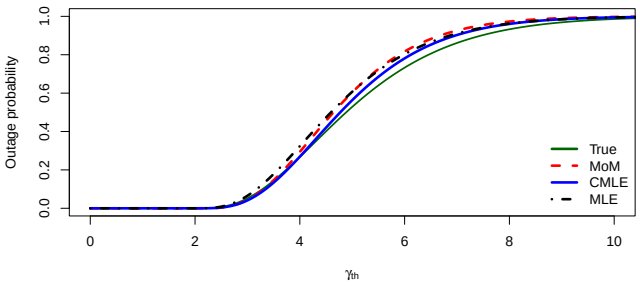



Fig. 2: True conditional outage probability $P_{\text{out}}(\gamma_{\text{th}} \mid z_1)$ and fitted curves from the MoM, CMLE, and MLE under dual-branch MRC ($n = 200$).

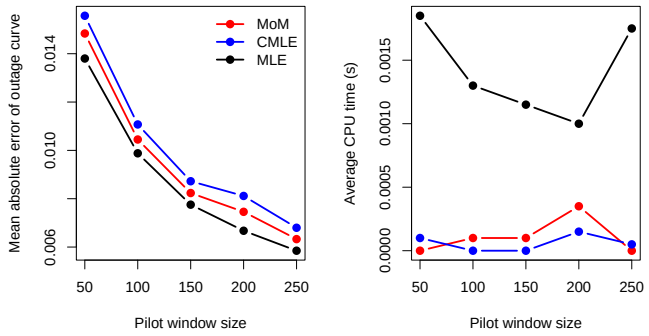



Fig. 3: Mean absolute error of the conditional outage curve (left) and average computation time (right), based on $R = 1000$ replications.

TABLE I: Mean absolute error of conditional-outage prediction under correct specification and misspecification ($n = 100$, $R = 1000$).

| Scenario | MoM | CMLE | MLE |
|---|---|---|---|
| Common scale | 0.0259 | 0.0294 | 0.0239 |
| Unequal power ($\beta_2/\beta_1 = 2$) | 0.0426 | 0.0536 | 0.0339 |
| Correlated branches ($\rho = 0.5$) | 0.0481 | 0.0776 | 0.0324 |

Figure 3 presents the Monte Carlo mean absolute error of the conditional outage curve and the average computation time. The error decreases with the pilot-window size for all estimators. The MLE has the smallest error, followed by the MoM and CMLE, while the CMLE computation time is approximately one order of magnitude below that of the MLE.

Table I uses the exact Gamma or Gaussian-copula conditional curve as reference. The observed first-branch SNR is fixed at its median, and the outage probabilities range from 0.01 to 0.99. The unequal-power case uses $\beta_2 = 2\beta_1$, whereas the correlated case uses equal Gamma marginals joined by a Gaussian copula with latent correlation $\rho = 0.5$. The fitted model retains the working common-scale and independent-increment assumptions. The errors increase under both departures. The MLE presents the smallest errors, while the CMLE is more affected by correlation. The largest Monte Carlo standard error among the reported MAEs is $1.0 \times 10^{-3}$.

For $L$ common-scale MRC branches, let $V_i \sim \text{Ga}(m_i, \beta)$ independently and $Z_k = \sum_{i=1}^{k} V_i$. The normalized increments have a Dirichlet distribution, and the remaining sum is independent of $Z_k$ with shape $M_{k+1:L} = \sum_{i=k+1}^{L} m_i$. Consequently,

$$P_{\text{out}}^{(L)}(\gamma_{\text{th}} \mid z_k) = \begin{cases} \dfrac{\gamma(M_{k+1:L}, (\gamma_{\text{th}} - z_k)/\beta)}{\Gamma(M_{k+1:L})}, & z_k < \gamma_{\text{th}}, \\ 0, & z_k \geq \gamma_{\text{th}}. \end{cases}$$

Selection and equal-gain combining are not SNR-additive and require different order-statistic or nonlinear-envelope constructions.

## VII. Conclusions

In this letter, we proposed a three-parameter bivariate Nakagami-$m$ model for branch-conditioned outage prediction in dual-branch MRC systems. A closed-form modified-likelihood estimator was derived, and conditions for consistency, asymptotic normality, and finite-sample feasibility were discussed. The simulations show that the CMLE provides a low-complexity alternative that improves on the MoM for $m_1$, while the MLE remains more accurate for $m_2$ and outage prediction. The robustness analysis quantified the effects of unequal powers and correlated branches, and the construction extends to common-scale multibranch MRC through Gamma partial sums. Reproducible code is available at https://github.com/ramospedroluiz/bivariatenakagami.

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